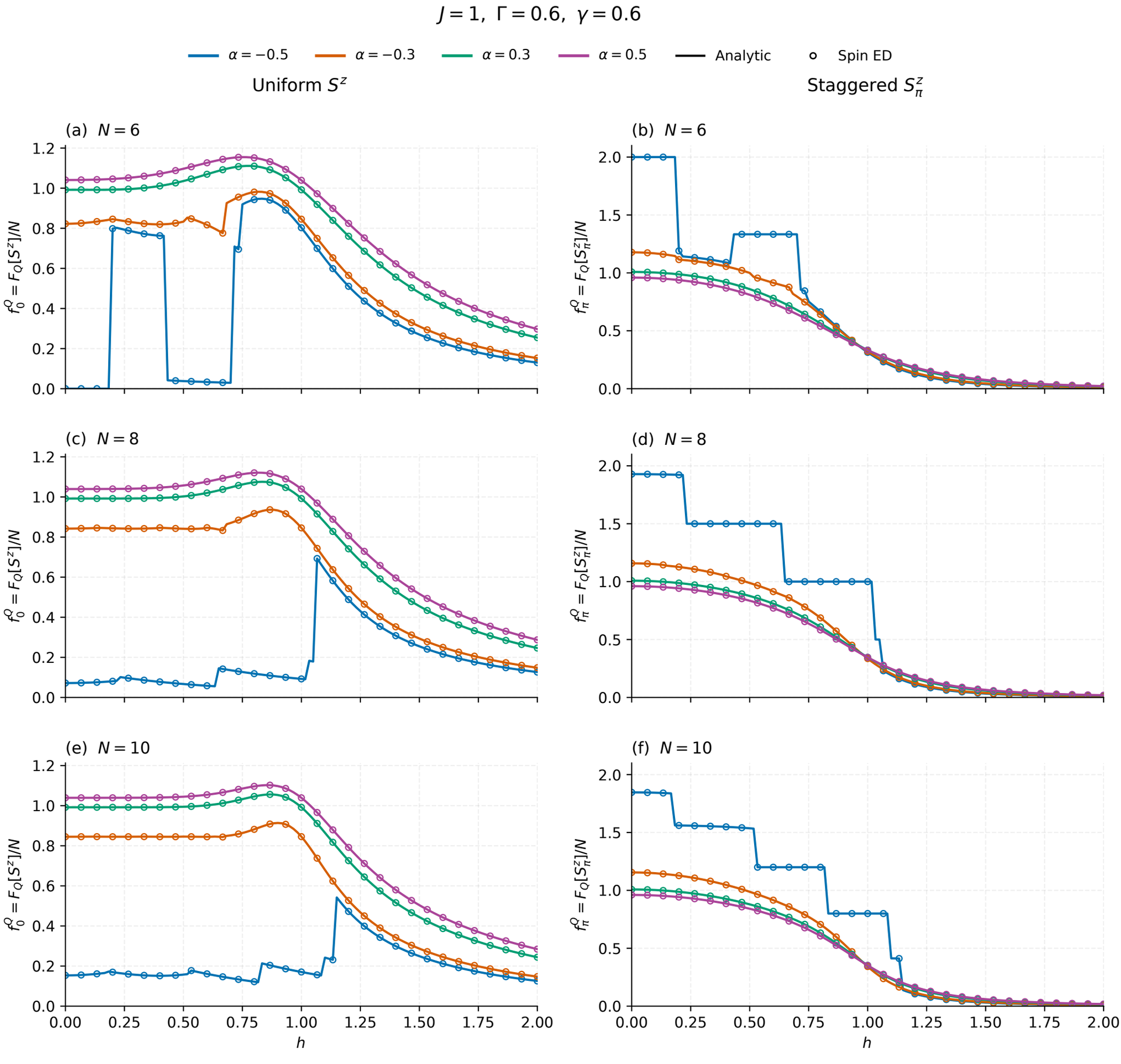

$J = 1,\ \Gamma = 0.6,\ \gamma = 0.6$
$\alpha = -0.5$
$\alpha = -0.3$
$\alpha = 0.3$
$\alpha = 0.5$
Analytic
Spin ED
Uniform S^z
Staggered S^z_π
(a) $N = 6$
(b) $N = 6$
(c) $N = 8$
(d) $N = 8$
(e) $N = 10$
(f) $N = 10$
$f_0^Q = F_Q[S^z]/N$
$f_\pi^Q = F_Q[S^z_\pi]/N$
h